CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting

A Capstone Proposal

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Chapter 1
Introduction and Background

1.1 Introduction

Technology is essential today as it drives innovation, enhances communication, and transforms education, industry, and society by providing efficient solutions to contemporary challenges. Digital transformation in education focuses on integrating digital tools and systems to improve teaching and learning processes. By using technologies such as online platforms, immersive environments, and interactive media, education becomes more adaptable, engaging, and supportive of higher-order thinking skills [1]. Learning Management Systems (LMS) enable teachers to deliver customized learning content, apply various teaching strategies, and engage students more effectively [2]. Embracing technology through digital transformation and academic systems underscores the vital role of innovation in shaping contemporary education and preparing learners for the demands of the modern world.

Despite all this digital transformation and inventing academic systems highlight the potential of technology to enhance teaching and learning, many schools still face challenges such as one of these is with traditional grading systems, where issues of fairness, consistency, and alignment with modern educational goals remain pressing concerns. These inefficiencies not only affect faculty productivity but also impact students, who may experience delays in accessing their grades and limited transparency in understanding their academic performance. Grades often fail to accurately reflect true learning and provide limited value as constructive feedback [3]. As educational systems continue to evolve, there is a need for more efficient and reliable solutions that can streamline grading processes and improve transparency in academic evaluation.

In recent years, the emergence of learning analytics and educational data mining has introduced new opportunities for improving academic systems. These approaches utilize student data to identify patterns and predict academic outcomes, enabling institutions to provide timely interventions. Techniques such as logistic regression have been widely used in predicting student performance due to their effectiveness in classification tasks. Studies highlight that learning analytics research still lacks comprehensive integration of multiple dimensions of student engagement, suggesting the need for more holistic approaches [4]. These developments emphasize the importance of integrating predictive capabilities into academic management systems.

Despite the growing adoption of LMS and predictive analytics, many educational institutions—particularly at the local level—have yet to implement integrated systems that combine grading management with academic performance forecasting. Existing systems often function only as repositories of grades rather than intelligent platforms capable of generating actionable insights. At the Candelaria

Institute of Technology in Cabadbaran Inc. (CITCI) College Department, the current grading process still presents challenges in terms of efficiency, accessibility, and effective use of academic data. These limitations highlight the need for a system that not only automates grading processes but also provides meaningful insights into student performance.

To address these challenges, this study proposes the development of **CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting**. The system aims to streamline grade encoding, automate General Weighted Average (GWA) computation, and generate accessible grade reports, while integrating a predictive model using logistic regression to forecast student academic performance. By combining grading automation with data-driven forecasting, the proposed system seeks to enhance administrative efficiency, improve transparency, and support proactive academic intervention, making it highly relevant in today's technology-driven educational environment.

1.2 Project Context

The operational efficiency of academic tracking stands as a critical factor in institutional management and student development. At the Candelaria Institute of Technology in Cabadbaran Inc. (CITCI) College Department, academic records and grading evaluations are handled through manual and semi-digital methods. Faculty members typically rely on localized, standalone electronic spreadsheets or manual ledger computations to input student marks, calculate raw scores, and determine final semestral standing. While functional on a basic level, this fragmented approach to record-keeping incurs significant administrative overhead, requiring repetitive manual entries that increase the likelihood of computation errors and processing delays.

This decentralized data management workflow directly creates an information bottleneck for both the faculty and the student body. Instructors must dedicate substantial time to formatting, recalculating, and compiling physical grade sheets, distracting from core instructional and mentoring responsibilities. Concurrently, students face prolonged wait times before obtaining verified insights into their academic progress or their definitive General Weighted Average (GWA). Because academic data remains localized within faculty files until the official end-of-term submissions, students operate within an informational blind spot, unable to continuously monitor their exact performance standings or receive structured feedback when it is most critically needed.

Beyond basic grade reporting, the lack of an integrated historical repository deprives the CITCI administration and academic heads of predictive operational intelligence. Under the current system architecture, identifying at-risk students who are likely to fail or drop out relies entirely on late-stage manual checks after final examinations are encoded. This structural delay prevents proactive, data-driven academic interventions during the active semester. To resolve these institutional

bottlenecks, there is an immediate need to transition toward a secure, centralized environment that automates grading computations while executing live performance forecasting through localized logistic regression, allowing CITCI to protect student data privacy while optimizing educational delivery.

1.3 Purpose and Description of the Project

The primary purpose of this project is to design and develop **CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting** for the Candelaria Institute of Technology in Cabadbaran Inc. (CITCI) College Department. This web-based platform serves to transition the institution from fragmented, spreadsheet-based grade computations toward a unified, automated academic management system. By consolidating student records into a secure repository, the system aims to eliminate human errors in calculations, decrease administrative delays for faculty members, and provide students with real-time, transparent access to their definitive academic standings and General Weighted Averages (GWA).

Architectural implementation of the proposed system centers on two core user categories: instructors and students. For faculty users, the platform introduces a streamlined **Grade Encoding Module** that automates grade entry and executes immediate, error-free **GWA Computations** aligned with the 1.00 to 5.00 numerical scale used by Philippine higher education institutions. This dashboard empowers instructors to generate downloadable, panel-ready grade sheets in standardized **PDF and Excel formats**. Conversely, the student-facing module provides an interactive interface where learners can track their running grade metrics per subject, view graphical data visualizations of their academic trajectories, and monitor progress across the active semester without depending on manual paper distributions or face-to-face administrative inquiries.

The defining operational feature of *CITCI GradeTrack* is its integrated **Academic Performance Forecasting Module**, which embeds predictive machine learning capabilities directly into the institutional grading workflow. Utilizing **Logistic Regression** as its core classification algorithm, the system evaluates structured grade parameters—specifically performance trends across Written Works, Performance Tasks, and mid-term examination marks—to calculate the probability of a student failing or dropping a course. Unlike generalized learning analytics platforms that rely on vague digital engagement metrics (such as login logs or page clicks), this forecasting architecture maps data strictly from actual assessment outcomes. This structural choice provides highly interpretable, real-time warning indicators, allowing academic heads and instructors to identify at-risk students early enough to deploy active, data-driven academic interventions.

To protect student credentials and institutional integrity, the system architecture is designed around rigid security and local privacy constraints. Developed utilizing a mobile-responsive interface to accommodate varying device availability among users, the application secures all transactions through strict role-based access control (RBAC). Furthermore, data management pipelines within the system are engineered to comply

with the mandates of the **Philippine Data Privacy Act of 2012 (RA 10173)**. By utilizing localized database deployment and incorporating strict masking protocols for dataset ingestion during model validation, *CITCI GradeTrack* delivers an optimized, low-cost administrative tool tailored explicitly to the curriculum and structural needs of the CITCI College Department.

1.4 Objectives of the Study

General Objective

To design, develop, and evaluate **CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting** for the CITCI College Department that integrates automated grading workflows, localized GWA tracking, and interpretable predictive capabilities using logistic regression, hosted locally to ensure complete compliance with the Philippine Data Privacy Act of 2012 (RA 10173).

Specific Objectives

- *To design the system architecture, ERD, and mobile-responsive user interfaces.*
- *To develop the instructor module for grade encoding, automated GWA computation, and PDF/Excel grade-sheet generation.*
- *To develop the student module for grade viewing, academic progress visualization, and personalized performance forecasts.*
- *To implement the academic performance forecasting using logistic regression.*
- *To test and evaluate the system using the ISO/IEC 25010 framework and the forecasting model using accuracy, precision, recall, and F1-score.*

1.5 Scope and Limitation of the Study

Scope

This study focuses on the design and development of **CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting** for the **Candelaria Institute of Technology in Cabadbaran Inc. (CITCI) College Department**. The system will cover the following features and modules:

- **Grade Encoding Module** – allows instructors to input student scores digitally and automate grade computation.
- **Automated GWA Computation** – calculates General Weighted Average (GWA) accurately and consistently.
- **Visual Grade Reports** – generates accessible grade sheets in PDF/Excel formats and provides charts for performance visualization.

- **Forecasting Module** – integrates logistic regression to predict student academic performance and identify at-risk learners.
- **User Access** – supports both instructors (grade encoding, report generation) and students (viewing grades, progress tracking).
- **Accessibility** – provides a mobile-responsive interface for convenient access to academic data.
- **Security** – ensures data confidentiality and integrity in compliance with the **Philippine Data Privacy Act of 2012 (RA 10173)**.

The system is intended to improve efficiency, transparency, and proactive academic intervention within the CITCI College Department.

Limitations

While the system aims to modernize grading and forecasting, it is subject to the following constraints:

- **Institution-Specific Scope** – the system is designed only for CITCI College Department and does not cover other institutions.
- **Forecasting Model** – limited to logistic regression; other advanced machine learning models (e.g., neural networks, ensemble methods) are not included.
- **Platform Support** – optimized for web and mobile browsers; no dedicated desktop or mobile application is developed.
- **Data Dependency** – forecasting accuracy depends on the completeness and quality of historical grade data provided by the institution.
- **Infrastructure Constraints** – performance may be affected by internet connectivity issues or limited hardware resources within the institution.
- **Budgetary and Time Limitations** – the system is developed within the available resources of the researchers and does not include large-scale deployment or commercial integration.

1.6 Significance of the Study

This study is significant because it addresses persistent challenges in academic grading and performance monitoring by introducing **CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting**. The project contributes to the modernization of educational processes and aligns with **Sustainable Development Goal (SDG) 9: Industry, Innovation, and Infrastructure**, which emphasizes building resilient infrastructure, promoting inclusive innovation, and fostering technological advancement in institutions. It also supports the **University Research Agenda on digital transformation in education**, ensuring that academic systems remain responsive to contemporary demands.

Target Beneficiaries

- **Students:** Students will benefit from timely access to their grades, automated GWA computation, and clear visual reports. The forecasting feature empowers them to monitor their academic progress, identify areas for improvement, and take proactive steps toward success. This enhances motivation, transparency, and ownership of their learning journey.

- **Teachers/Instructors:** Faculty members will experience reduced workload and improved efficiency through automated grade encoding and error-free computation. The system provides reliable reports, minimizes manual recalculation, and supports data-driven feedback for students. This allows instructors to focus more on teaching and mentoring rather than administrative tasks.

- **Academic Administrators/Department Heads:** Administrators will gain access to consolidated grade reports and performance forecasts, enabling better decision-making in academic planning, resource allocation, and student support programs. The system strengthens institutional accountability and ensures compliance with standards such as the **Philippine Data Privacy Act of 2012 (RA 10173)**.

- **Institution (CITCI College Department):** The institution benefits from modernized grading infrastructure that enhances credibility, efficiency, and competitiveness. By adopting innovative academic systems, CITCI positions itself as a forward-looking institution aligned with global educational trends.

- **Community and Educational Sector:** At a broader level, the project contributes to the advancement of educational technology in the Philippines. It serves as a model for integrating grading management with predictive analytics, encouraging other institutions to adopt similar innovations that improve transparency, efficiency, and student outcomes.

CHAPTER 2
REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the related literature and systems relevant to the development of CITCI GradeTrack: A **Grading Management System with Academic Performance Forecasting for the CITCI College Department.** The review begins with foundational studies on **digital transformation in education** and **academic performance monitoring**, proceeds through discussions on **challenges in traditional grading syste**ms and **predictive analytics using logistic regression**, and ends with a review of comparable **existing grading and learning management systems**. It also identifies the research gap that highlights the need for an integrated system that combines automated grade computation with academic performance forecasting and data-driven decision support.

Review of Related Literature

2.1 Digital Transformation in Education

Digital transformation in education leverages modern technologies such as smart devices, learning applications, interactive content, distance learning, and immersive tools like virtual and augmented reality to create innovative environments that foster critical thinking, analytical skills, and more engaging learning experiences [1]. Similar study, highlighted that adaptive learning technologies allow for personalized instruction, enabling students to learn at their own pace while improving outcome [4]. In the Philippine context, reported that digital platforms enhanced student motivation and participation among senior high school learners, while stressed the importance of accessible online systems to sustain engagement during remote learning [5] Artificial intelligence further enhances learning by enabling adaptive and intelligent educational systems [6]. Blended learning has been found to significantly enhance academic performance, self-directed learning skills, and student attitudes. A quasi-experimental study comparing traditional and blended approaches reported higher academic achievement in the blended learning group ($p = 0.001$), along with improved interaction and engagement. These findings support the effectiveness of blended models in modern education [7].

A study conducted at Chinhoyi University of Technology examined the effectiveness of virtual e-learning platforms as a response to the COVID-19 pandemic. Using structural equation modeling on data from undergraduate students, the findings revealed that e- learning adoption was significantly hindered by factors such as the adoption of e-learning was hindered by infrastructural and economic barriers, including inadequate access to devices, unstable internet connectivity, and high technological costs [8]. Mobile learning

applications have been shown to significantly improve skill development among learners.

A quasi-experimental study involving out-of-school youth in the Philippines found that participants who used mobile learning apps performed significantly better

than those using traditional module-based instruction ($p < 0.05$). The results highlight the effectiveness of mobile learning as a flexible and accessible approach for enhancing both practical and cognitive skills [9]. The reviewed studies show that digital platforms improve academic performance, engagement, and accessibility, but challenges such as limited resources and connectivity remain. These findings support the need for a grading management system that ensures accurate, accessible, and timely academic records, while integrating performance forecasting to identify at-risk students and enable early intervention.

2.2 Academic Performance Monitoring

Monitoring student academic performance is a critical component of educational systems, as it provides insights into learning progress and supports informed decision-making. Progress monitoring is a structured process where teachers set goals, regularly assess student performance, and adjust instruction to ensure milestones are met, providing targeted support—especially for learners with individualized needs—and fostering inclusive, data- driven academic success [10]. Continuous monitoring of student performance supports the development of systems like *CITCI GradeTrack*, as it enables automated tracking, accurate grade computation, and data-driven academic performance forecasting to improve decision-making and student outcomes.

2.3 Challenges in Traditional Grading Systems

Research consistently highlights inefficiencies, fairness concerns, and transparency issues in manual grading practices, emphasizing that conventional grading often fails to provide constructive feedback, limiting its usefulness in guiding student improvement [3]. Grades provide so little meaningful information to students that the time and effort spent assigning, receiving, and interpreting them often outweighs their value, and even in academic settings where they are expected to be most useful, grades tend to obscure learning rather than clarify it [11]. Traditional grading systems often lead to grade inflation, fixed mindsets, inconsistency across instructors, and discourage risk- taking, prompting educators to explore fairer alternatives that emphasize mastery, personalized feedback, and intrinsic motivation [12]. Most Philippine colleges use a numerical grading system on a 1.00 to 5.00 scale where 1.00 is the highest grade (Excellent) and 5.00 is failing. The minimum passing grade is 3.00, corresponding to 75 percent performance. Overall academic performance is measured by GWA (General Weighted Average), computed by multiplying each subject's grade by its credit units and dividing by total units enrolled [13]. This system, while standardized, still suffers from the same inefficiencies, lack of meaningful feedback, and inconsistencies noted in traditional grading practices.

These limitations reinforce the need for innovative solutions like CITCI GradeTrack, which not only automates grade computation but also integrates forecasting and analytics to provide more transparent and actionable insights into student performance. The limitations of traditional grading—such as inefficiency, lack of

meaningful feedback, and inconsistency—highlight the need for systems like *CITCI GradeTrack*, which can automate grading processes, improve transparency, and support data-driven insights to make academic evaluation more accurate and meaningful for student development.

2.4 Predictive Analytics and Logistic Regression for Student Performance

Educational data mining techniques have demonstrated high accuracy in predicting academic performance when learner-activity data are available [14]. A study introduced a predictive-analytics framework that supports student retention through early alerts, while earlier learning-analytics research confirmed that engagement and performance data can be used to detect academic risk and trigger timely interventions before withdrawal occurs [15] [16]. Logistic regression has been widely used in predicting student outcomes due to its interpretability and effectiveness in classification tasks [17]. Student dropout remains a major challenge in higher education, particularly in developing regions. A study conducted at Mindanao State University utilized multiple machine learning algorithms to predict at-risk students using historical academic and demographic data. The results showed that advanced models such as Extreme Gradient Boosting achieved high predictive accuracy (over 90%), demonstrating the effectiveness of data-driven approaches in identifying students at risk of dropping out [18].

Early warning systems have been developed to monitor student academic performance and provide timely intervention. One study implemented an automated mechanism integrated into an academic system, where students receive alerts when their grades fall below a set threshold (e.g., ≤60). The system demonstrated high accuracy and positive user feedback, helping students maintain performance and enabling proactive academic support. These findings highlight the effectiveness of automated monitoring and notification systems in improving student outcomes [19]. Studies comparing machine learning models and logistic regression have shown that advanced models such as gradient-boosted decision trees and random forests can achieve slightly higher predictive accuracy in complex datasets. However, findings also indicate that the performance gap becomes minimal as the number and quality of predictor variables increase.

Logistic regression remains competitive, particularly due to its simplicity, interpretability, and efficiency, making it a practical choice for predictive modeling in structured data environments [20]. The reviewed studies show that student data can be used to accurately predict academic performance and identify at-risk learners. Predictive models such as logistic regression, along with early warning systems, enable timely intervention and improve student retention. These findings support the development of a grading management system that integrates performance forecasting and automated monitoring to enhance academic decision-making.

2.5 Data Privacy and Ethical Use of Academic Data

A Study emphasized that the use of digital technologies in education must be

guided by principles of fairness, transparency, and accountability [21]. Similar studies also stated, highlighted the privacy risks associated with handling student data, particularly the potential misuse of sensitive academic information [22]. In the context of the proposed system, these concerns translate into key design considerations such as secure data storage, role-based access control, and compliance with the Philippine Data Privacy Act of 2012 (RA 10173). Additionally, the use of performance forecasting requires transparency in how student data is analyzed to ensure ethical and responsible academic decision-making.

2.6 Automated Grading and Feedback

AI-driven grading and feedback are reshaping education by improving speed, consistency, and personalization in assessment. While not a replacement for teachers, these systems enhance how educators understand and support students. The future of education lies not only in faster grading but in deeper understanding, supported by intelligent systems that enable educators to act on students' needs more effectively [23]. Assessment and feedback in higher education must evolve to promote learning, ensure rigour, and highlight employability. Programmatic assessment (PA), first developed in medical education and now adopted across professional fields, offers a solution. It is guided by eight core principles and supported by four practical steps for programme teams, drawn from Dutch experiences. This approach also carries implications for leaders and support staff, ultimately fostering a more constructive and effective assessment culture in higher education [24]. Platforms like Edundy illustrate connected, analytics- driven approaches, while systems such as CITCI GradeTrack extend this potential by embedding forecasting and academic management, ensuring efficient assessment and deeper understanding of student performance and outcomes. This demonstrates that while Automated grading and feedback enhance speed, consistency, and personalization, systems like CITCI GradeTrack extend this potential by integrating forecasting and academic management features, ensuring not only efficient assessment but also deeper insights into student performance and outcomes.

Review of Related Systems

2.7 Google Classroom

The study surveyed 292 college students at Northern Negros State College of Science and Technology to evaluate Google Classroom's effectiveness, using data mining algorithms (Random Forest, C4.5, Naive Bayes) with various validation techniques to compare performance metrics and recommend the best approach for improving student engagement [25]. The findings show that applying machine learning algorithms to student survey data can identify effective strategies for improving engagement, guiding systems like CITCI GradeTrack to integrate data- driven insights for more responsive and impactful learning experiences.

2.8 Moodle

Educational institutions are increasingly encouraged to utilize data from learning management systems to improve teaching and learning outcomes. A study using Moodle data examined whether student activity logs could predict academic performance in a blended learning environment. The findings showed that models using complete course data were able to predict student grades with moderate accuracy and pass/fail outcomes with high accuracy. However, predictions made using early-term data were less reliable, though performance improved when more weeks of data were included. The study suggests that LMS activity data can support the development of early-warning systems for identifying at-risk students [26]. While LMS-based models can predict student performance, they are limited by delayed accuracy and reliance on activity data; *CITCI GradeTrack* addresses these gaps by integrating grading data with logistic regression to provide more reliable and timely academic performance forecasting within a unified grading management system.

2.9 Canvas LMS

A study using Canvas LMS explored how learning analytics can be used to predict student academic performance. Using multiple regression analysis, the study examined behavioral indicators such as assignment submission, page views, and participation time to determine their influence on final grades . The findings suggest that LMS data can help identify patterns related to student success and support timely academic interventions. However, the study also noted limitations, particularly that LMS data does not capture important factors such as socio-emotional and contextual influences, making it only one component in predicting student performance [27]. While Canvas-based analytics can predict student performance using behavioral data, it remains limited and disconnected from grading processes; *CITCI GradeTrack* addresses these gaps by integrating grading data with logistic regression to provide a more structured, system-based academic performance forecasting approach.

2.10 Blackboard Learning Management System

A study examined the relationship between student engagement in the Blackboard Learning Management System and academic performance among engineering students. Using statistical analysis, the findings revealed a positive correlation between the time students spent on the platform and their final examination grades. The study also observed that students tend to engage more during specific periods, indicating structured learning behavior. These results suggest that student interaction data from LMS platforms can serve as valuable indicators of academic performance and support data-driven monitoring systems [28]. While Blackboard demonstrates that student engagement is related to academic performance, it lacks predictive capabilities and proactive intervention features. The proposed system addresses this gap by integrating grading management with performance forecasting to support early identification of at-risk

students.

2.11 Schoology Learning Management System

Schoology is a learning management system that enables educators to create lesson plans, manage course content, and administer assessments such as quizzes, assignments, and exams. It also supports classroom management features including attendance tracking, grading, and student performance monitoring. Additionally, the platform provides communication tools such as discussion forums, allowing interaction among teachers, students, and parents, as well as a professional learning network for educators [29]. While Schoology offers comprehensive tools for content delivery and grading management, it primarily focuses on instructional support and lacks built-in predictive capabilities for forecasting student academic performance. This limitation highlights the need for systems that integrate grading with performance forecasting, as proposed in this study.

2.12 Automated Grading System

Automated Grading Systems (AGSs) have been explored in higher education, particularly in computer science, to streamline assessment and reduce teacher workload. While AGSs promise efficiency and improved practices, studies reveal challenges such as tensions and breakages in assessment processes, showing that even advanced systems require careful integration to ensure fairness and effectiveness [30]. This shows that while Automated Grading Systems improve efficiency, their limitations in fairness and integration highlight the need for CITCI GradeTrack, which fills these gaps by automating grade computation, ensuring transparency, and adding forecasting features to provide deeper, data- driven insights into student performance.

2.13 Comparative Analysis of Existing Systems

The proposed CITCI GradeTrack system is benchmarked against the reviewed LMS platforms across auto- grading, predictive at- risk detection, local hosting compliance, curriculum alignment, cost, and setup complexity. The comparison highlights the unique capabilities contributed by the proposed system— such as built-in forecasting, RA 10173- aligned hosting, and customization for CITCI's curriculum— that are not natively available in leading alternatives like Google Classroom.

Feature	Google Classroom	Moodle	Canvas LMS	Blackboard LMS	Schoology LMS	Proposed CITCI Grade Track
Auto-Grading (Objective Items)	Limited	Plugin Required	Limited	Limited	Limited	Yes
Predictive at-risk detection	No	Yes (Activity, logs, delayed)	Yes (Regression Based)	Correlation Only	No	Yes (Logistic Proactive alerts
Local hosting (RA 10173-aligned)	No	No	No	No	No	Yes (Privacy compliant hosting)

Tailored to CITCI curriculum	No	No	No	No	No	Yes (Customized Design)
Cost	Free	Free / open-source	Medium	High	Medium	Free/open-source
Setup complexity	Low	Medium	Medium	High	Medium	Medium (Streamlined)

Table 1. Comparative analysis of existing Grading System and Forecasting systems versus the proposed Grade Track System with Student performance Forecasting.

Unlike existing generic platforms, the principal contribution of this study is a focused integration of automated grading, error-free GWA computation, and predictive at-risk detection using logistic regression—combined with RA 10173 - aligned local hosting and tailoring to the CITCI curriculum. Bundled into a single, low- cost platform, the proposed system is designed specifically for the needs of a small Philippine higher education institution.

2.14 Research Gap

From the reviewed literature and systems, 9 specific gaps motivate this study:

• **Grading systems are not centralized and automated.** Many institutions still rely on manual or semi-digital grading processes, leading to inefficiencies, inconsistencies, and delays in grade computation and reporting.

• **Predictive analytics is not integrated into grading workflows.** Existing systems treat prediction as a separate feature rather than embedding it directly into grading systems for real-time academic monitoring.

• **Reliance on activity-based data limits prediction accuracy.** Current LMS platforms depend heavily on engagement metrics (e.g., clicks, submissions), which may not accurately reflect true academic performance, especially in early stages.

• **Delayed identification of at-risk students.** Many systems can only provide reliable predictions after sufficient data is collected, reducing the effectiveness of early intervention strategies.

• **Lack of interpretable and practical predictive models.** Advanced machine learning models offer high accuracy but are often complex and difficult to implement, while simpler models like logistic regression are underutilized in actual systems.

• **Limited feedback and transparency in grading.** Traditional and existing digital grading systems often fail to provide meaningful insights, making it difficult for students to understand and improve their performance.

• **Fragmentation between grading, monitoring, and analytics.** Existing

platforms separate grade management, performance tracking, and analytics, resulting in inefficient and disconnected academic evaluation processes.

• **Lack of context-specific systems for local institutions.** Most available systems are generalized and not tailored to institutional grading standards such as the Philippine GWA system used in CITCI.

• **Data privacy and ethical considerations are not fully embedded.** While literature highlights privacy concerns, many systems do not fully integrate secure, role-based, and locally compliant data management practices.

This study addresses these gaps by proposing CITCI GradeTrack, a Grading Management System that integrates automated grade computation, logistic regression–based academic performance forecasting, and real-time monitoring within a single platform. The system provides transparent, interpretable, and data-driven insights while ensuring secure and locally compliant data handling, specifically tailored for the CITCI College Department.

2.15 Synthesis of the Review

The reviewed literature and systems collectively demonstrate that digital transformation, learning management systems, and predictive analytics have significantly improved educational delivery, student engagement, and academic monitoring. Studies highlight that LMS platforms such as Google Classroom, Moodle, Canvas, Blackboard, and Schoology effectively support content management and student interaction, while predictive analytics techniques enable the identification of at-risk students and support early intervention. However, these systems remain limited due to their reliance on activity-based data, delayed prediction accuracy, and lack of integration with actual grading processes. Furthermore, traditional grading systems continue to suffer from inefficiency, inconsistency, and limited feedback, reducing their effectiveness in guiding student improvement. Although automated grading and machine learning models enhance efficiency and predictive capabilities, many approaches are either too complex or not directly embedded into institutional grading workflows.

Based on these gaps, the proposed system, CITCI GradeTrack, integrates automated grade management with logistic regression–based academic performance forecasting to provide a more accurate, transparent, and interpretable approach to student evaluation. By combining grading data with predictive analytics in a single platform, the system enables real-time monitoring, early identification of at-risk students, and data-driven academic decision-making, while also addressing issues of usability, fairness, and data privacy within the specific context of the CITCI College Department.

2.16 Theoretical Framework

The development of CITCI GradeTrack: A Grading Management System with

Academic Performance Forecasting is anchored on established theories in information systems and educational data analysis. These theories provide the foundation for understanding user adoption, data-driven academic monitoring, and predictive modeling within the proposed system.

The Technology Acceptance Model (TAM) explains how users come to accept and use a technology based on two key factors: perceived usefulness and perceived ease of use. In the context of this study, the model supports the design of CITCI GradeTrack as a system that enhances grading efficiency while remaining user-friendly for both instructors and students. By ensuring that the system is easy to navigate and beneficial in managing academic records, it increases the likelihood of adoption within the institution.

In addition, the concept of Learning Analytics provides a theoretical basis for utilizing student data to monitor academic progress and support decision-making. Learning analytics emphasizes the importance of analyzing educational data to identify patterns, track performance, and improve learning outcomes. This supports the system's capability to generate insights from grading data and assist educators in identifying students who may require academic intervention.

Furthermore, the study is grounded in Logistic Regression, a statistical method widely used for predicting outcomes based on historical data. Logistic regression is particularly suitable for forecasting student performance, such as determining whether a student is at risk or likely to pass, due to its interpretability and efficiency. Its application in CITCI GradeTrack enables the system to provide clear and data-driven predictions that can support timely and informed academic decisions.

Collectively, these theories establish a strong foundation for the proposed system by linking user acceptance, data utilization, and predictive modeling. They justify the integration of automated grading and academic performance forecasting in CITCI GradeTrack, ensuring that the system is not only functional but also theoretically sound and aligned with current educational and technological practices.

2.17 Conceptual Framework

This study follows an **Input–Process–Output (IPO) model**, which serves as the blueprint for the development of CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting. The framework illustrates how input data are transformed through system processes into meaningful outputs that support academic decision-making.

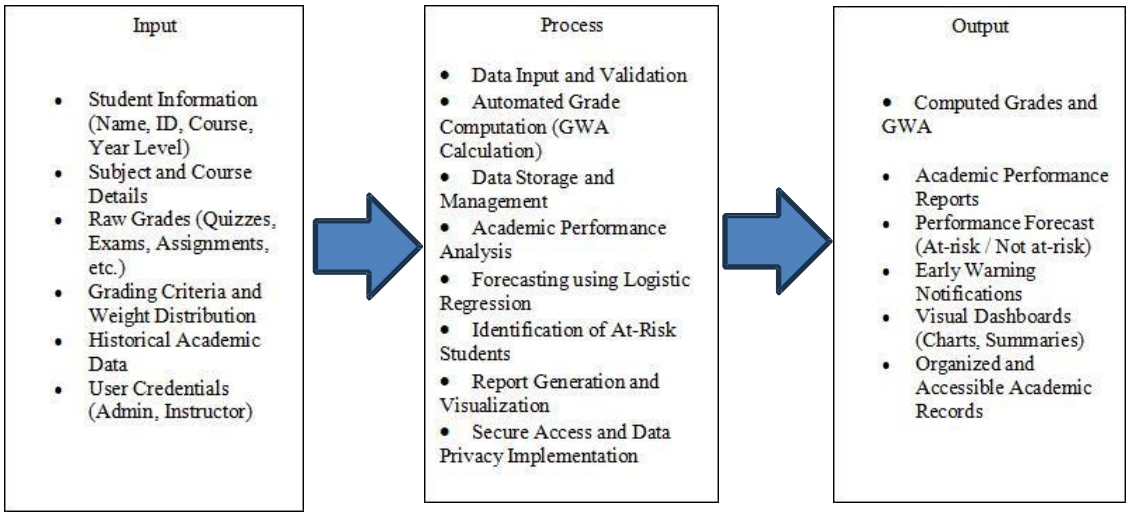


Figure 1. Conceptual Framework of the Study.

The conceptual framework of the study is based on the Input–Process–Output (IPO) model, which illustrates how CITCI GradeTrack transforms academic data into meaningful insights. The input stage includes student information, subject details, grading components, grading criteria, and historical academic records, which serve as the foundation of the system. In the process stage, the system performs data validation, automated grade computation, data storage, and academic performance analysis, followed by forecasting using Logistic Regression to identify at-risk students. It also generates reports, visualizations, and ensures secure access through data privacy measures. The output stage presents the results in the form of computed grades, General Weighted Average (GWA), performance forecasts, early warning notifications, and organized academic records. This framework demonstrates how the system integrates grading management and predictive analytics to support accurate evaluation, timely intervention, and data-driven academic decision-making within the CITCI College Department.

2.18 Technical Background

The proposed system is a web-based grading management system with integrated academic performance forecasting designed for the CITCI College Department. The frontend is developed using HTML, CSS, and JavaScript, providing a responsive and accessible user interface that allows instructors and students to view grades, academic summaries, and performance forecasts through any modern web browser. The backend is implemented using PHP with the Laravel framework, which manages authentication, grade processing workflows, request routing, and communication between the application and the database. Persistent data storage is handled using MySQL, a relational database management system that stores student records, subject details, computed grades, General Weighted Average (GWA), and academic performance prediction results.

The forecasting module is implemented using Python integrated with Scikit-learn, focusing on a lightweight predictive analytics approach. The core component utilizes Logistic Regression to predict student academic performance based on computed grades, historical academic records, and performance indicators such as assessment scores and overall grade trends. This model is used to classify students as either at-risk or not at-risk, enabling early academic intervention. The Python-based prediction service is exposed as an API that communicates with the Laravel backend through HTTP requests, allowing the forecasting logic to operate independently from the main system.

The overall system follows a client-server architecture where the web application serves as the main platform for grade management, while the Python-based analytics module functions as a separate service for academic performance forecasting. This modular design ensures scalability, maintainability, and the ability to improve or retrain the predictive model without affecting the core grading system.

CHAPTER 3
METHODOLOGY

This chapter presents the blueprint of the project. It describes research design, system architecture and requirements, predictive analytics approach, statistical treatment of evaluation data, development phases, and testing and ethical considerations guiding the development of CITCI GradeTrack: A Grading Management System with Academic Performance Forecasting.

3.1 Research Design

This study employs developmental and descriptive research design. The developmental design is used to design and build the proposed CITCI GradeTrack system, while the descriptive design is used to evaluate the system's usability, functionality, and effectiveness as perceived by instructors and students of the CITCI College Department. The system is developed using an Agile Software Development approach, which allows iterative development through repeated cycles of planning, implementation, testing, and evaluation based on stakeholder feedback.

3.2 Project Planning

In this phase, the researchers identified key academic challenges within the CITCI College Department, particularly the inefficiencies of manual grading, delays in grade computation, lack of centralized academic records, and limited tools for identifying at-risk students. Initial requirements were gathered through interviews with selected instructors and surveys distributed to students to understand grading practices and academic monitoring needs.

A project plan was developed that includes system objectives, scope, technology stack, development timeline (represented through a Gantt chart), and required hardware and software resources. A feasibility analysis was also conducted:

- **Technical Feasibility** – The chosen technologies (Laravel, MySQL, PHP, and Python with Scikit-learn) are stable, well-documented, and capable of supporting grading management and predictive analytics.
- **Operational Feasibility** – Instructors and students have access to internet-enabled devices suitable for browser-based system use.
- **Economic Feasibility** – The system utilizes open-source technologies, making it cost-effective and suitable for institutional deployment without additional licensing costs.

3.3 Requirements Specification

Requirements were gathered through (a) interviews with faculty members of the CITCI College Department, (b) student surveys regarding academic monitoring needs and grading concerns, and (c) analysis of current manual grading and

record-keeping processes.

3.4 Functional Requirements

- Allow instructors to input, manage, and compute student grades efficiently.
- Enable students to access learning materials, submit assignments, and take quizzes.
- Automatically compute grades based on predefined grading criteria and weight distribution.
- Generate academic performance reports for instructors and administrators.
- Predict at-risk students based on academic performance data using forecasting techniques.
- Provide early warning notifications for students with declining performance.
- Offer analytics dashboards summarizing student performance trends.
- Ensure secure access for instructors, students, and administrators through authentication.

3.5 Non-Functional Requirements

- The system must be accessible through modern web browsers without installation.
- The system must respond to user requests within 2 seconds under normal usage conditions.
- Secure authentication must be implemented using hashed passwords and encrypted sessions.
- The system must support concurrent users within a small-to-medium academic department (50–500 users).
- Academic records must be securely stored in compliance with the Philippine Data Privacy Act of 2012 (RA 10173).
- Predictive output must be interpretable, showing contributing academic factors for each prediction result.

3.6 Hardware Requirements

Developers (Development Environment):

- Processor: Intel Core i5 or AMD Ryzen 5 (or higher)
- RAM: 8 GB minimum (16 GB recommended)
- Storage: 256 GB SSD or higher
- Internet: Stable broadband connection (10 Mbps or higher)

End Users (Students and Instructors):

- Device: Laptop, desktop, tablet, or smartphone
- Processor: Dual-core or higher
- RAM: 4 GB minimum
- Internet: At least 5 Mbps connection

Server Requirements:

- Processor: Quad-core or higher
- RAM: 8–16 GB
- Storage: 500 GB SSD or higher
- Network: Stable high-speed connection for concurrent access

3.7 Software Requirements

- Operating System: Windows 10/11, Ubuntu Linux, or macOS
- Web Server: Apache or Nginx
- Backend: PHP 8.x with Laravel Framework
- Database: MySQL 8.x
- Frontend: HTML5, CSS3, JavaScript
- AI/Prediction Module: Python 3.x with Scikit-learn, Pandas, and NumPy
- Version Control: Git
- Local Development Environment: XAMPP or Laragon

3.8 System Models and Architecture

The system design is represented through three major diagrams. The Data Flow Diagram illustrates how grading data moves from instructors to the system and is processed into computed grades and performance forecasts. The Entity Relationship Diagram defines the relationships between key entities such as students, subjects, grades, assessments, and prediction outputs. The System Architecture Diagram illustrates the layered structure of CITCI GradeTrack, consisting of the frontend interface, Laravel backend for grade management, Python-based forecasting module using Logistic Regression, and MySQL database for secure data storage and retrieval.

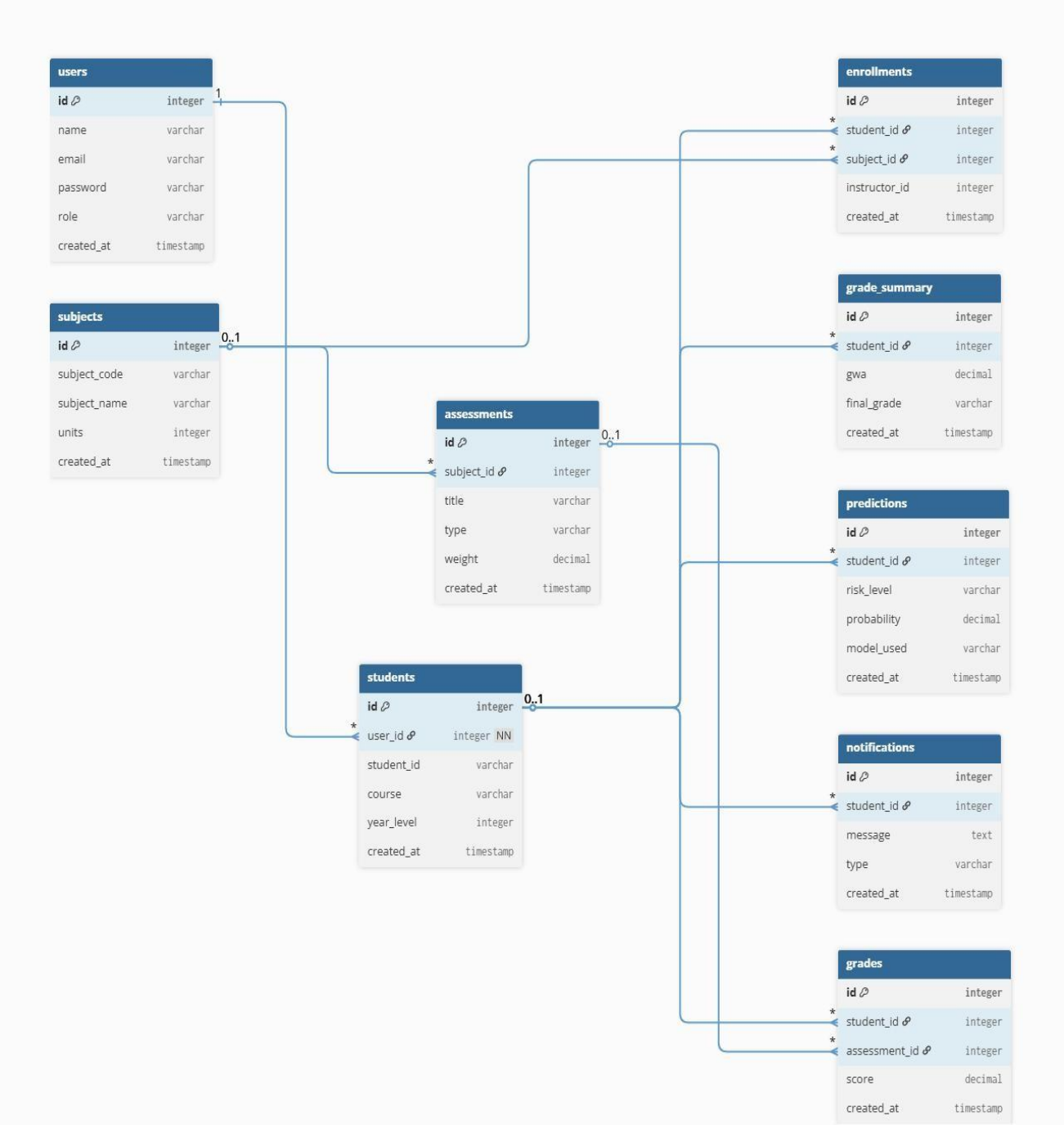


Figure 3. Entity Relationship Diagram (ERD) of the Proposed CITCI GradeTrack System.

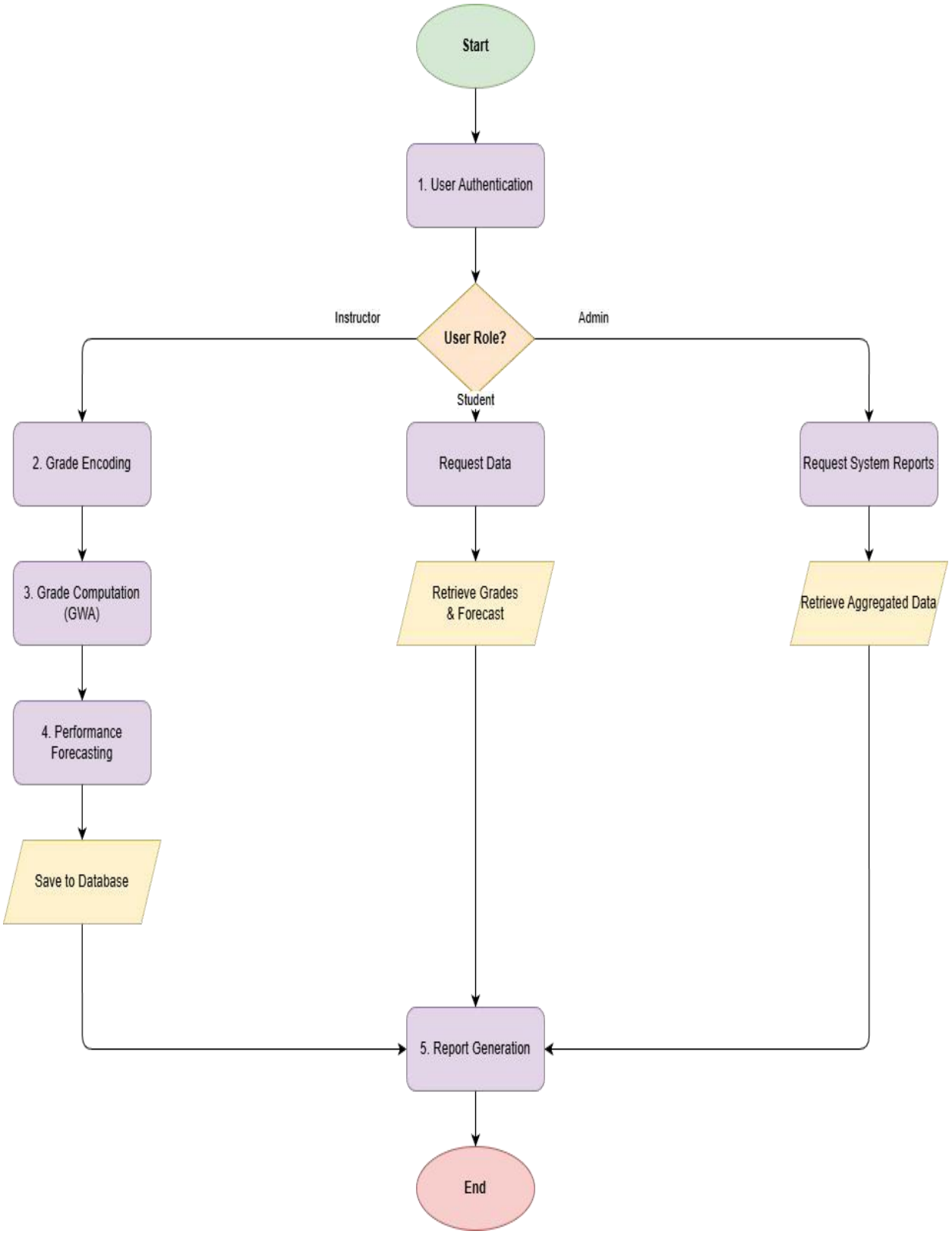


Figure 4. Data Flow Diagram of the Proposed CITCI GradeTrack System.

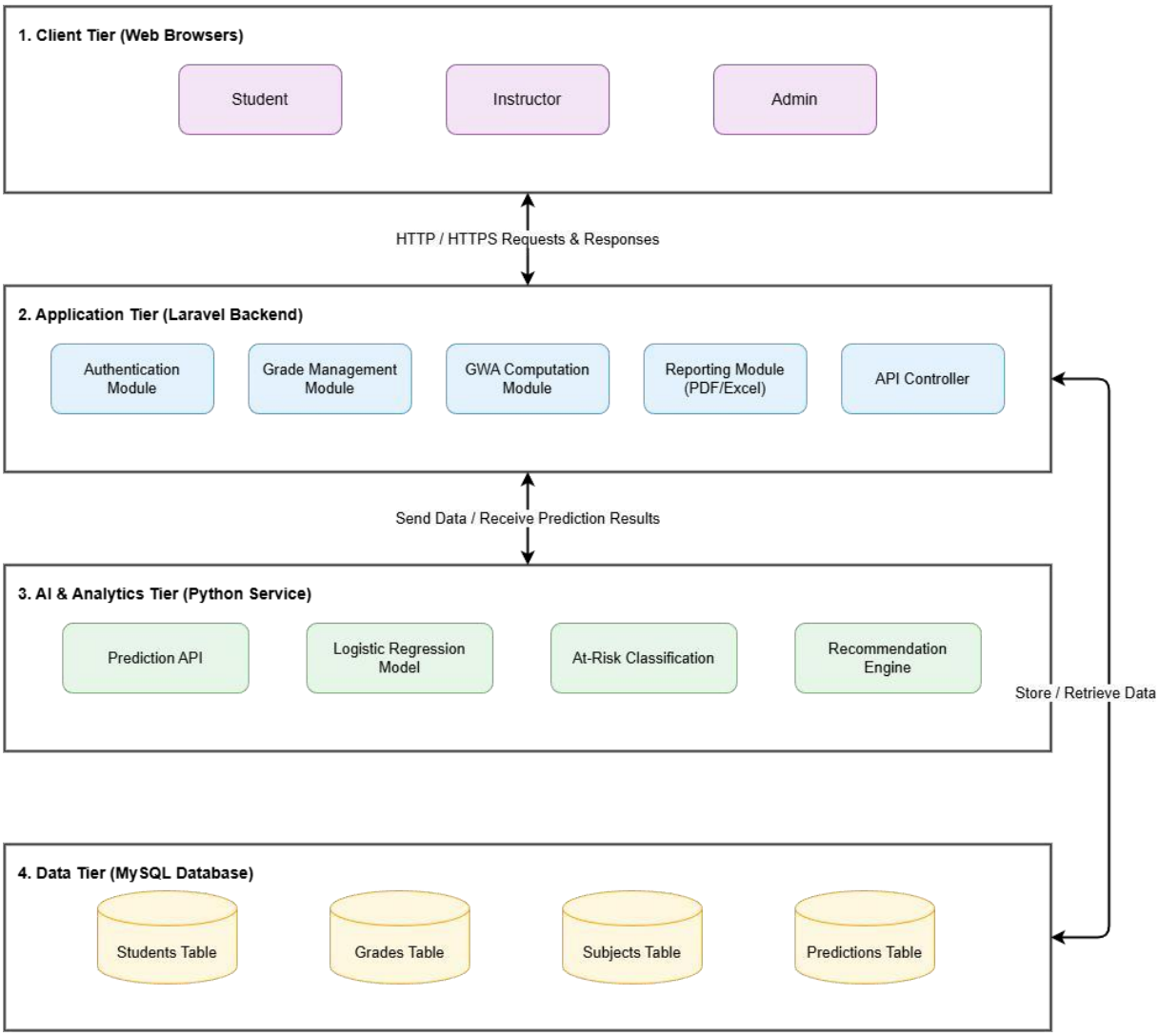


Figure 5. System Architecture of the Proposed CITCI GradeTrack System.

The system architecture is organized into three tiers. The **client tier** represents the users—students, instructors, and administrators—who interact with the system through a web interface. Their requests and responses are transmitted via HTTP/HTTPS. The **application tier** contains the main functional modules, including authentication and role management, grade management, GWA computation, reporting, and an API controller that coordinates communication with other components. The **AI and analytics tier** processes academic data through a prediction API, logistic regression model, at- risk classification, and recommendation engine, generating insights and forecasts that are sent back to the application tier. At the foundation, the **data tier** serves as the persistent storage, holding records such as student information, grades, subjects, and prediction results, ensuring reliable data exchange across all layers.

3.9 AI Algorithms

Three lightweight AI components are integrated into the system. Each is chosen to be implementable by an undergraduate team within a single capstone term while still being defensible as machine learning rather than a heuristic rule.

3.9.1 At-Risk Student Classifier (Logistic Regression)

A binary logistic regression model is used to classify whether a student is at risk of failing a course. The model utilizes key academic performance indicators as input features, including quiz score average (x_1), submission timeliness rate (x_2), login frequency over the past two weeks (x_3), and assignment completion rate (x_4). These variables are used to estimate the probability of a student being at risk, expressed using the logistic function:

$$P(\text{at-risk} \mid x) = 1 / (1 + \exp(-z)),$$

$$\text{where } z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

A student is classified as “at-risk” when the predicted probability exceeds a threshold value of 0.7. The model is trained using historical academic performance data from the CITCI College Department, and in the absence of sufficient initial datasets, a synthetic dataset based on institutional performance patterns is used for initial deployment. Model parameters are estimated using maximum likelihood estimation through the Scikit-learn implementation of Logistic Regression. To ensure interpretability, each prediction includes contributing feature weights, allowing instructors to understand the factors influencing the classification and supporting transparent academic decision-making.

3.10 Statistical Treatment of Data

The data gathered from the system evaluation will be analyzed using descriptive statistical methods in accordance with ISO/IEC 25010 software quality standards. These statistical tools will be used to measure the acceptability, usability, and effectiveness of the CITCI GradeTrack system as perceived by the respondents.

(1) **Weighted Mean.** The weighted mean will be used to determine the overall rating of each evaluation item. It is computed using the formula:

$$WM = \Sigma fx / \Sigma f,$$

where f represents the frequency of responses and x represents the corresponding Likert scale value.

(2) Likert Scale Interpretation.

Range	Interpretation
4.21 – 5.00	Excellent
3.41 – 4.20	Very Good
2.61 – 3.40	Good
1.81 – 2.60	Fair
1.00 – 1.80	Poor

(3) Percentage Distribution.

$$P = (f / N) \times 100,$$

where N is the total number of respondents.

(4) Standard Deviation.

To capture rating variability across respondents:

$$SD = \sqrt{(\Sigma f(x - WM)^2 / (N - 1))}$$

(5) Prediction-Model Metrics.

To evaluate the performance of the academic forecasting model, standard classification metrics are used to assess the effectiveness of the logistic regression classifier in identifying at-risk students. These metrics ensure that the predictive component of CITCI GradeTrack produces reliable and meaningful results.

The model is evaluated using accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of predictions, while precision determines the proportion of correctly identified at-risk students among all predicted at-risk cases. Recall is particularly important in this study as it measures the model's ability to correctly identify actual at-risk students, supporting early intervention. The F1-score provides a balanced measure between precision and recall to ensure overall model reliability.

These evaluation metrics are used to validate the performance of the forecasting model and ensure that it provides accurate, interpretable, and useful predictions for academic decision-making within the CITCI College Department.

3.11 Project Development Phases

The system is developed using the Agile methodology, which allows iterative and incremental development through structured sprints. Each sprint focuses on specific system modules to ensure continuous improvement, integration, and evaluation based on stakeholder feedback.

- **Sprint 1: User Interface and Authentication Module**
  Development of the system interface and core access control features, including login, registration, and role-based access for administrators, instructors, and students.
- **Sprint 2: Academic Structure Management Module**
  Implementation of course, subject, and instructor management functionalities, allowing proper organization of academic offerings within the system.
- **Sprint 3: Assessment and Grading Module**
  Development of quiz creation, assignment submission, and automated grading for objective assessments, forming the core of the grading system.
- **Sprint 4: Student Performance Monitoring Module**
  Implementation of student dashboards for viewing grades, General Weighted Average (GWA), and academic progress tracking.
- **Sprint 5: Predictive Analytics and AI Features**
  Integration of the academic forecasting system using Logistic Regression for at-risk student classification, along with additional features such as analytics dashboards and performance-based insights for instructors.
- **Sprint 6: Testing, Deployment, and Evaluation**
  System testing, debugging, deployment preparation, and conduct of User Acceptance Testing (UAT) with selected CITCI instructors and students to evaluate system usability and functionality.

3.12 Operational Framework

• User authentication. Instructors, students, and administrators sign in using assigned credentials, with passwords hashed and sessions secured using HTTPS.
• Course and material management. Instructors upload course materials, create assignments, and configure quizzes within their assigned subjects.
• Student access. Students view materials, submit assignments, and take quizzes for the subjects in which they are enrolled.
• Assessment and grading. Objective assessments are automatically graded, while subjective outputs are evaluated by instructors using rubrics.
• AI recommendations and risk detection. The system analyzes student performance to generate study recommendations and identify at-risk students for early intervention.

• Learning analytics. Instructors access dashboards showing student performance, grades, and at-risk indicators.
• Data storage. All academic records and system data are stored in a MySQL database hosted on department-controlled infrastructure.

3.13 Testing and Evaluation

The deployed system is evaluated through four complementary methods. Functional testing verifies core features such as authentication, grade computation, performance forecasting, and data processing work as intended. Performance testing measures system response time, stability, and handling of concurrent users. User Acceptance Testing (UAT) is conducted with selected CITCI instructors and students using a Likert-scale questionnaire based on the ISO/IEC 25010 software quality model. Predictive model evaluation measures accuracy, precision, recall, and F1-score of the logistic regression classifier in identifying at- risk students. The evaluation focuses on usability, functionality, efficiency, and reliability of the system.

3.14 Data Analysis

Quantitative data from User Acceptance Testing are analyzed using the statistical methods described in Section 3.10. Likert-scale responses are summarized using weighted mean and standard deviation based on ISO/IEC 25010 quality attributes. Percentage distribution is used to describe response frequencies per evaluation item. Where applicable, one-way ANOVA is used to compare responses across different user groups. For the forecasting model, performance is evaluated using accuracy, precision, recall, and F1-score, with confidence intervals generated through bootstrap resampling of the test dataset.

3.15 Ethical Considerations

• Data privacy and confidentiality of all respondents and academic records are strictly maintained throughout the study.

• Informed consent is obtained from all participating instructors and students prior to system evaluation and data collection.

• Authorization is secured from the CITCI College Department before system deployment and testing activities are conducted.

• All student data and academic records are stored securely in compliance with the Philippine Data Privacy Act of 2012 (RA 10173).

• AI outputs are designed to be explainable, where every at-risk prediction includes the contributing input factors to ensure transparency and accountability.

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